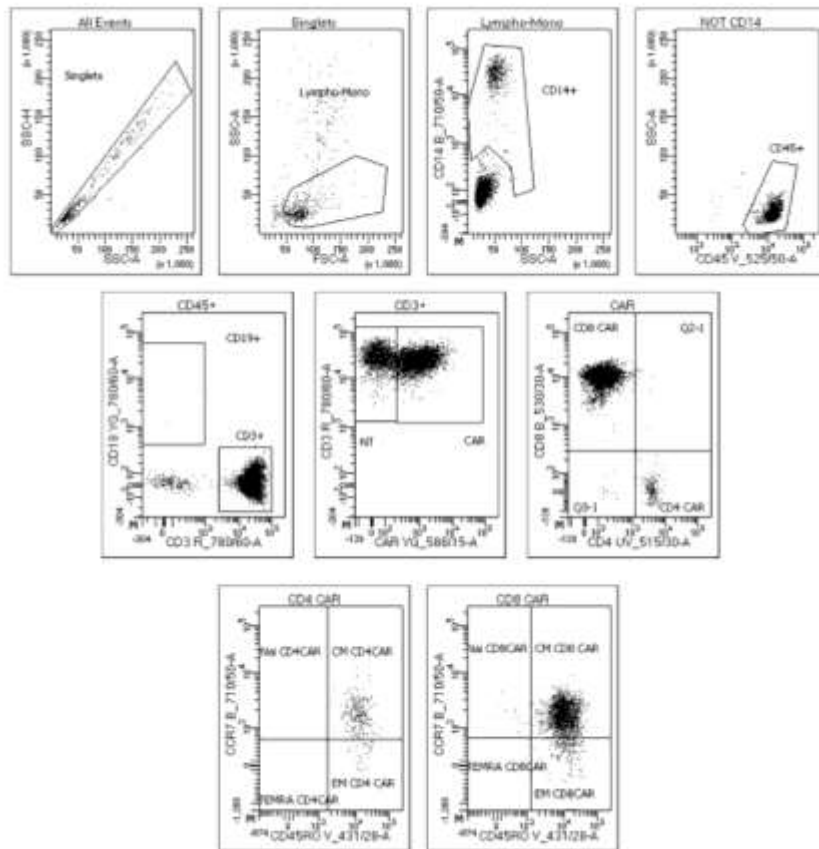

Donor-derived GD2-specific CAR T cells in relapsed or refractory neuroblastoma

In the format provided by the
authors and unedited



Supplemental Figure 1. Gating strategy. The gating strategy for the analysis of CAR+ T cells was carried out considering the following steps:

- 1) Singlets gating using SSC-H and SSC-A scatters
- 2) in the Singlet gate the lympho-monocytes populations are placed using SSC-A vs FSC-A
- 3) the CD14+ cells are excluded inverting the gate
- 4) in order to exclude the contamination of reticulocytes, the CD45+ cells are gated
- 5) in the CD45+ cells are plotted CD19+ and CD3+ lymphocytes
- 6) the CAR+ and Not-transduced cells are gated in the CD3+ total population
- 7) in the CAR+ gate the different subpopulations are distinguished using CD4 and CD8 parameters
- 8) the memory subpopulations in the CD4+ CAR+ and CD8+ CAR+ cells are characterised using CCR7 and CD45RO expression

NOTE : The hierarchy is indicated at the top of each plot

 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
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Background

Neuroblastoma (NBL) represents the most common extracranial solid tumor of childhood (8-10% of all malignant tumors) and accounts for about 12-15% of all cancer-related deaths in pediatric patients. NBLs are highly heterogeneous neoplasms that vary in terms of location, histological, and biological characteristics. NBL is characterized by a broad spectrum of clinical behavior, ranging from spontaneous regression or maturation into a benign ganglioneuroma to highly aggressive disease with metastatic spread. The extreme clinical heterogeneity is linked to various clinical and biological factors, including the patient's age, tumor stage, histology, and genetic and chromosomal abnormalities. Despite recent advances, the treatment of high-risk NBL remains one of the major challenges in pediatric oncology, with long-term survival rates below 50% (1). Although the current therapeutic strategy has prolonged event-free survival (EFS), neither European nor North American cooperative group studies published to date have demonstrated a significant improvement in overall survival (OS) (2).

In light of this, new therapeutic approaches are urgently needed. Among the therapies developed so far, the importance of cell therapy with GD2-CAR T cells emerges. Treatment with autologous T cells genetically modified to express a chimeric antigen receptor (CAR) targeting the GD2 molecule has shown extremely promising results in the treatment of patients with high-risk, relapsed, or refractory neuroblastoma within our Phase I/II academic trial (NCT03373097) (3,4). The results obtained from the first 27 patients treated in the GD2CAR01 protocol, open at our center, showed a treatment response rate of 63%, with 9/27 patients (33.3%) achieving complete remission of the disease and 8/27 (29.6%) achieving partial remission—a particularly relevant result given the disease characteristics of the treated patients (3,4).

Despite the clinical results obtained, some limitations of the CAR T-cell approach have emerged over time. In particular, patients who relapse early after receiving highly lymphodepleting chemotherapy cycles or, even more significantly, after a hematopoietic stem cell transplant (HSCT), present with markedly reduced circulating lymphocyte counts, making it impossible to collect the necessary number of cells to initiate the production of autologous CAR T cells (5). Furthermore, even if the

 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
---	---

absolute count were compatible with collection, the fitness of these cells is certainly reduced due to the treatments received (6). These factors, in some cases, make the production of autologous CAR T cells impossible. To overcome these limitations, the production of allogeneic CAR-T cells from donor cells is under study and development in several centres worldwide.

The main additional risk associated with using an allogeneic product, compared to an autologous one, is the possibility of alloreactive activation of CAR T cells, leading to GvHD. However, it is important to note that the presence of the iC9 suicide gene in our product significantly mitigates this risk. Several scientific studies have demonstrated that the use of donor lymphocytes (DLI), genetically modified to express iC9 as a safety switch in case of GvHD, and infused in patients undergoing allogeneic bone marrow transplantation for malignant or benign hematologic diseases, has a high safety profile (7).

Safety switch: inducible caspase 9 (iC9)

The inducible caspase 9 (iC9) suicide gene is a human caspase engineered at the binding domain, replaced with that of the human FK506-binding protein (FKBP). The resulting chimeric caspase remains inactive within the cells until a dimerizing agent, AP1903, is administered (8). This agent is otherwise inert and non-toxic but induces dimerization of the FKBP domain, leading to the activation of the apoptotic cascade and ultimately, cell apoptosis. Introducing the suicide gene into CAR cells adds an additional safety level, offering the option to activate it in the event of excessive or uncontrollable toxicity. iC9 has already been extensively tested in clinical settings, both at our center and others, proving to be safe and effective, capable of rapidly eliminating the cells that express it even in vivo (9-12).

In our experience with autologous CAR-T products in the context of ongoing academic trials at our center, when necessary to control grade 4 CRS, the activation of the suicide gene resulted in rapid elimination of CAR-T cells, which was associated with a corresponding clinical response [personal communication from Prof. Locatelli].

Criteria for patient inclusion

- Diagnosis of HR-NBL with advanced disease, who failed the first line and at least 2 additional lines of therapy AND

 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
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- Profound lymphopenia preventing collection of autologous T cells AND/OR
- Failure of a previous treatment with autologous GD2-CART01
- Absence of alternative therapeutic options
- Recover from the toxic effect of previous chemotherapies: grade 4 and or 3 non-hematologic toxicities must have resolved to grade ≤ 2 ; if some effects of the therapies have become chronic (i.e. treatment associated thrombocytopenia), the patient must be clinically stable, according to the opinion of the treating physicians, and meet all other eligibility criteria.
- Age: 12 months – 25 years.
- Voluntary informed consent is given. For subjects < 18 years old their legal guardian must give informed consent. Pediatric subjects will be included in age appropriate discussion and verbal assent will be obtained for those greater than or equal to 12 years of age, when appropriate.
- Clinical performance status: Patients > 16 years of age: Karnofsky greater than or equal to 60%; Patients less than or equal to 16 years of age: Lansky scale greater than or equal to 60%.

Criteria for patient exclusion

- Pregnant or lactating women
- Severe, uncontrolled active intercurrent infections, active hepatitis B or hepatitis C infection, HIV infection
- History of grade 3 or 4 hypersensitivity to murine protein-containing products or to GD2-CART01
- Hepatic function: Inadequate liver function defined as total bilirubin > 4x upper limit of normal (ULN) or transaminase (ALT and AST) > 6 x ULN based on age and laboratory specific normal ranges
- Renal function: serum creatinine > 3x ULN for age.
- Blood oxygen saturation < 90%.
- Cardiac function: Left ventricular ejection fraction lower than 45% by ECHO.
- Marrow function: ANC lower than 500/mm³ and/or platelets lower than 20.000 (not reached by transfusion).
- Congestive heart failure, cardiac arrhythmia, psychiatric illness, or social situations that would limit compliance with study requirements or in the opinion of the PI would pose an unacceptable risk to the subject.

 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
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- Untreated CNS metastasis; patients with previous CNS tumor involvement that has been treated and is stable for at least 6 weeks following completion of therapy are eligible.
- Concurrent or recent prior therapies, before infusion:
 - a. Systemic steroids (at a dose equivalent to or greater 2 mg/kg prednisone) in the 2 weeks before infusion. Recent or current use of inhaled/topical/non-absorbable steroids is not exclusionary.
 - b. Systemic chemotherapy in the 2 weeks preceding infusion.
 - c. Immunosuppressive agents less than or equal to 30 days prior to infusion.
 - d. Radiation therapy must have been completed at least 3 weeks prior to infusion.
 - e. I131-MIBG therapy must have been completed at least 6 weeks prior to infusion
 - f. Anti-GD2 murine monoclonal antibody (ch14.18 antibody) in the 2 weeks preceding infusion
 - g. Other anti-neoplastic investigational agents currently or within 30 days prior to infusion;
 - h. Exceptions:
 - ii. Subjects receiving steroid therapy at physiologic replacement doses only are allowed provided there has been no increase in dose for at least 2 weeks prior to infusion

Rationale for dose of ALLO_GD2-CART01

The proposed individual therapeutic use concerns the treatment of a pediatric patient with multiple relapses of NBL, with persistent disease after multiple lines of therapy. We propose to treat the child with allogeneic CAR-T cells expanded from mononuclear cells obtained from an HLA-compatible intrafamily donor.

In the literature and from our group's experience, autologous GD2-CAR-T cells have been used at doses of up to 10×10^6 CAR⁺ cells/kg body weight. In the study by Brudno and colleagues on allogeneic CD19-CAR T cells produced from HLA-matched donors, the cohort of patients treated with the highest cell dose received up to 7.8×10^6 CAR⁺ cells/kg body weight, with no identified

 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
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correlation between the administered dose and the occurrence of toxicity (13). Based on this evidence, we have selected a dose of 3×10^6 /kg CAR+ cells/kg body weight, which, with the same autologous product, has proven effective in our academic trial and is believed to represent the best balance between efficacy and toxicity risk.

Protocol of use

The administration of ALLO_GD2-CART01 will follow a conventional lymphodepletion chemotherapy regimen consisting of fludarabine and cyclophosphamide, administered for 3 consecutive days, followed by 2 days of rest to allow for drug washout.

The allogeneic CAR-T cells will be administered as an intravenous infusion through a central venous catheter or peripheral access.

The infusion will be performed following the established clinical practice at our centre for autologous products. Premedication with chlorpheniramine and ondansetron will be administered and the cell infusion will last 10-30 minutes. Vital parameters will be monitored during the infusion and for 2 hours afterward.

The patient will receive the drug product under inpatient care and will remain admitted until resolution of the treatment-related acute toxicities.

Clinical and laboratory monitoring

- Before lymphodepleting chemotherapy, the patient will undergo screening tests, including blood count, liver and kidney function, respiratory and cardiac function, disease status (via bone marrow aspirate and diagnostic lumbar puncture), microbiological screening for HIV, HBV, HCV, and exclusion of active infections.
- Daily clinical evaluations
- Daily laboratory tests (blood gas analysis, blood count, chemistry, coagulation).
- Monitoring for any signs of GvHD (skin rash, diarrhea and its quantification, liver function, fever).
- Monitoring for any signs of toxicity (cytokine release syndrome, neurotoxicity, cytopenia, hypogammaglobulinemia).

 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
---	---

- Monitoring of circulating CAR-T cell levels.
- Evaluation of the response to the therapy via bone marrow aspirate, CT scan and MIBG-scintigraphy on week +6 after infusion (subsequent checks will be scheduled based on clinical progression, as per usual clinical practice).
- Monitoring for RCR (up to 15 years post-infusion).

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 Bambino Gesù OSPEDALE PEDIATRICO	Allogeneic anti-GD2 Chimeric Antigen Receptor-Expressing T cells in a pediatric patient affected by r/r HR-NBL – hospital exemption research protocol
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